



Name: Ameen UL Husnaat

Roll No: 2025-SET-05

Course Title: Software Configuration Mangement Technology

Course Code: ST-1206

Class: BS SET 2ND Semester (2025-2029)

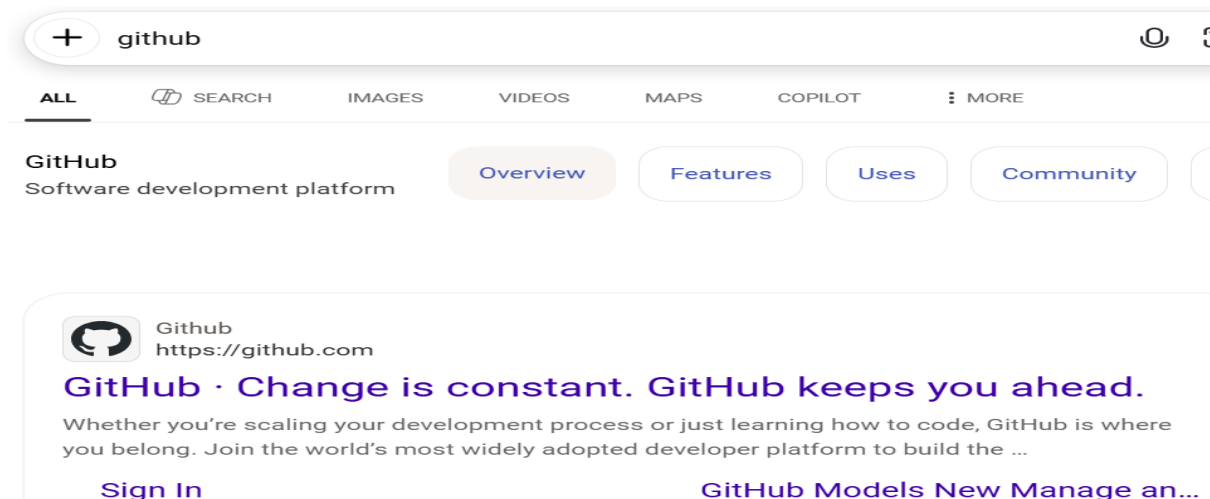
Submitted To: Sir Fahad Nisar

Department: Software Engineering

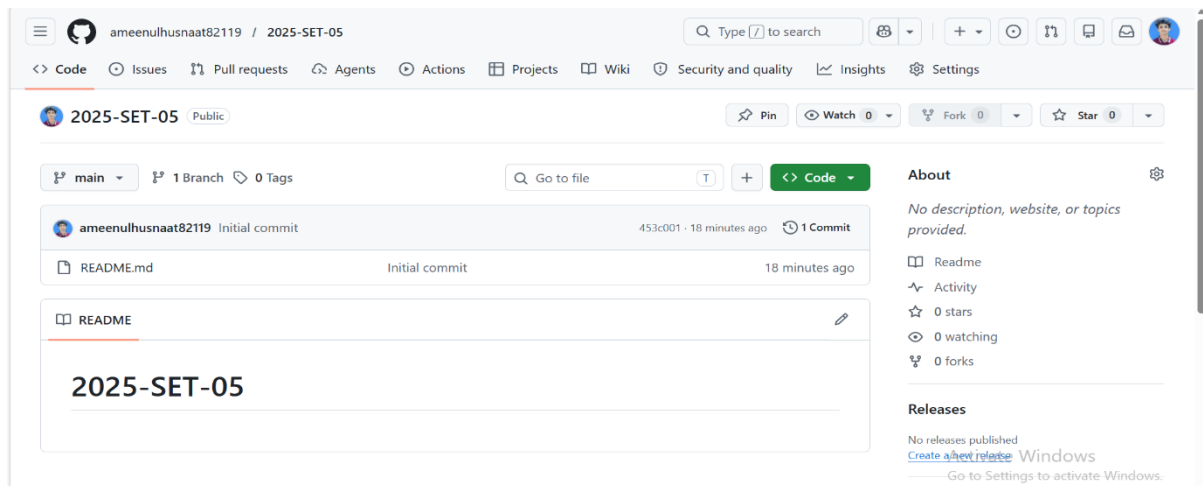
Lab Tasks 5,6,7 & 8

Lab Task 5 # Create a GitHub Repository & Cloning a Public Repository

Step 1: Create a GitHub Account



Step 2: Create a Public Repository to a local folder



Step 3: Connect Remote Repository to a local Folder

```

PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> git init
Initialized empty Git repository in D:/husnaat/University work/2nd semester/SCMT_Lab_Tasks/2025-SET-05/.git/
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> git remote
add origin https://github.com/Ameen-UL-Husnaat/scm-lab.git
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> git remote
-v
origin https://github.com/Ameen-UL-Husnaat/scm-lab.git (fetch)
origin https://github.com/Ameen-UL-Husnaat/scm-lab.git (push)
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> █

```

Step 4: Clone a Public GitHub Repository

```

PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> git clone
https://github.com/ameenulhusnaat82119/Lab-Project.git
Cloning into 'Lab-Project'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05> cd Lab-Pro
ject
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git log --oneline
12449a8 (HEAD -> main, origin/main, origin/HEAD) Add files via upload
4bc28ba Initial commit
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> █

```

Lab Task 6 # Pushing Code to Remote Repository (Main & Feature Branch)

Step 1: Push code to Main Branch

```

> git add .
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git commit -m "Add main-file.txt"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git branch -M main
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git push -u origin main
info: please complete authentication in your browser...
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> █

```

Step 2: Create a new branch push code

```
> git checkout -b feature-branch
Switched to a new branch 'feature-branch'
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> echo "This is a feature file" > feature-file.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git add .
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git commit -m "Add feature-file.txt on feature-branch"
[feature-branch 3b2b0e1] Add feature-file.txt on feature-branch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature-file.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> git push -u origin feature-branch
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 343 bytes | 343.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-branch' on GitHub by visiting:
remote:   https://github.com/ameenulhusnaat82119/Lab-Project/pull/new/feature-branch
remote:
To https://github.com/ameenulhusnaat82119/Lab-Project.git
 * [new branch]      feature-branch -> feature-branch
branch 'feature-branch' set up to track 'origin/feature-branch'.
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
```

Step 3.1: Verify both Branches are on Github

The screenshot shows the GitHub interface for a repository named 'Lab-Project'. At the top, it indicates the repository is 'Public' and has 0 watchers. A yellow banner states 'feature-branch had recent pushes 5 minutes ago' with a 'Compare & pull request' button. Below this, the branch 'feature-branch' is selected, showing it is 2 branches and 0 tags away from the main branch. A status bar indicates 'This branch is 1 commit ahead of main'. The commit history shows three commits: 'Electricity Bill Calculator.cpp' (5 months ago), 'README.md' (5 months ago), and 'feature-file.txt' (6 minutes ago). The 'feature-file.txt' commit is highlighted, showing its message 'Add feature-file.txt on feature-branch' and hash '3b2b0e1'. A 'Contribute' button is visible at the bottom right.

Step 3.2:

The screenshot shows the GitHub repository page for 'Lab-Project', which is public. At the top, there's a notification bar indicating that the 'feature-branch' had recent pushes 6 minutes ago, with a 'Compare & pull request' button. Below this, the repository navigation bar shows 'main' as the selected branch, with '2 Branches' and '0 Tags'. A search bar 'Go to file' and a '+ Code' button are also present. The commit history table lists two commits by user 'ameenulhusnaat82119': 'Electricity Bill Calculator.cpp' (5 months ago) and 'README.md' (5 months ago). Below the commit history, the 'README' file is selected, showing the title 'Lab-Project'.

Lab Task 7 # Creating & Merging Pull Request on GitHub

Step 1: Open a pull request

The screenshot shows the 'Create pull request' form in GitHub. The top bar indicates the base branch is 'main' and the compare branch is 'feature-branch', with a status 'Able to merge'. The 'Add a title' field contains 'Merge feature-branch into main'. The 'Add a description' field has a text area with the content 'we are merging feature-branch into main and create pull request of repository Lab Project'. On the right side, there are sections for 'Reviewers' (No reviews), 'Assignees' (No one—assign yourself), 'Labels' (None yet), 'Projects' (None yet), and 'Milestone' (No milestone). At the bottom right, there's a 'Create pull request' button and a link to 'Activate Windows'.

Step 2: Review for Merge Conflict

 **Open**

Merge feature-branch into main #1
ameenulhusnaat82119 wants to merge 1 commit into `main` from `feature-branch` 

  Add feature-file.txt on feature-branch 3b2b0e1





No conflicts with base branch
Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)



Add a comment

Write

Preview

H B I   <> 

Add your comment here...

Step 3: Merge the Pull Request

 **Merged**

Merge feature-branch into main #1
ameenulhusnaat82119 merged 1 commit into `main` from `feature-branch` 

  Add feature-file.txt on feature-branch 3b2b0e1

  ameenulhusnaat82119 merged commit **19020d2** into `main` now

Revert



Pull request successfully merged and closed

You're all set — the `feature-branch` branch can be safely deleted.

Delete branch



Add a comment

Write

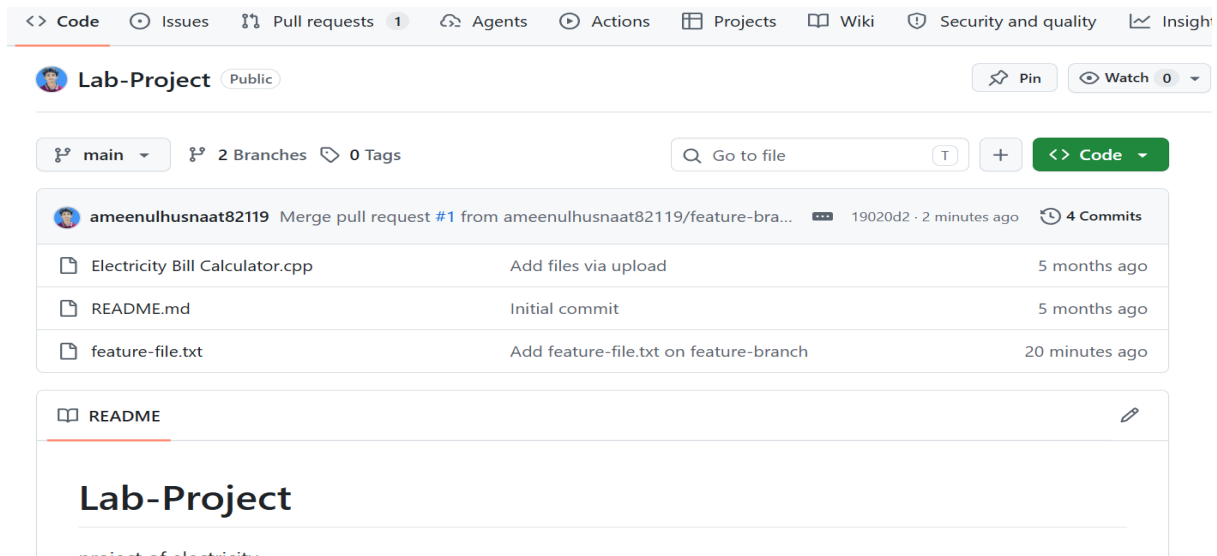
Preview

H B I   <> 

Add your comment here...

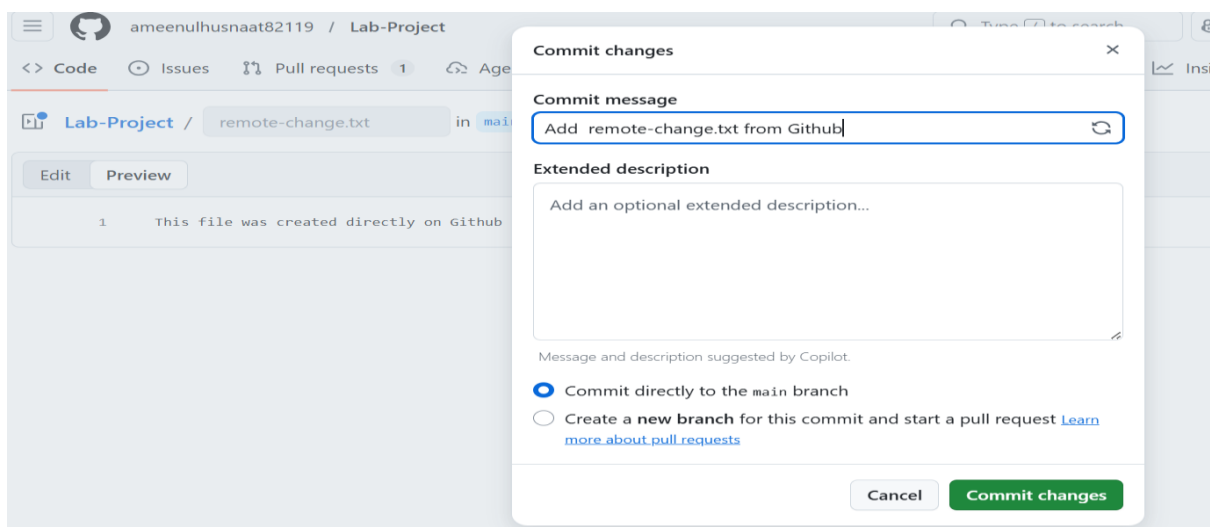
 Markdown is supported  Paste, drop, or click to add files

Step 4: Verify the Merge



Lab Task 8 # Pulling Remote Changes to Your Local Repository

Step 1: Make a Change Directly on GitHub



Step 2: Check Your Local Repository Before Pulling

```
> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> █
```

Step 3: Pull the Remote Changes

```
> git pull origin main
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.83 KiB | 93.00 KiB/s, done.
From https://github.com/ameenulhusnaat82119/Lab-Project
* branch          main          -> FETCH_HEAD
   12449a8..cb63517  main        -> origin/main
Updating 12449a8..cb63517
Fast-forward
 feature-file.txt | Bin 0 -> 50 bytes
 remote-change.txt | 1 +
 2 files changed, 1 insertion(+)
 create mode 100644 feature-file.txt
 create mode 100644 remote-change.txt
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> █
```

Step 4: Verify the Change Locally

```
> git log --oneline
cb63517 (HEAD -> main, origin/main, origin/HEAD) Add remote-change.txt from Github
19020d2 Merge pull request #1 from ameenulhusnaat82119/feature-branch
3b2b0e1 (origin/feature-branch, feature-branch) Add feature-file.txt on feature-b
branch
12449a8 Add files via upload
4bc28ba Initial commit
PS D:\husnaat\University work\2nd semester\SCMT_Lab_Tasks\2025-SET-05\Lab-Project
> █
```

_____ THE END _____